Test Cases Document

Agent Based Vision-Mixer for Video Broadcasts

Final Year Project

Contents

1 Module 1: Vehicle Detection **2**
- 1.1 TC-VD-01: Detect Vehicle in Clear Frame . . . 2
- 1.2 TC-VD-02: No Detection on Empty Frame . . . 2
- 1.3 TC-VD-03: Detect Multiple Vehicles Simultaneously . . . 3
- 1.4 TC-VD-04: Detection Under Low Light Conditions . . . 3
- 1.5 TC-VD-05: Filter Non-Vehicle Objects . . . 4

2 Module 2: Camera Switching Engine **5**
- 2.1 TC-CS-01: Switch to Higher Resolution Camera . . . 5
- 2.2 TC-CS-02: Hysteresis Prevents Unnecessary Switching . . . 5
- 2.3 TC-CS-03: Smooth Transition on Camera Switch . . . 6
- 2.4 TC-CS-04: Handle Overlap Zone Between Two Cameras . . . 6
- 2.5 TC-CS-05: Fallback When Active Camera Loses Vehicle . . . 7

3 Module 3: Re-Identification / Cross-Camera Matching **8**
- 3.1 TC-RI-01: Assign Global ID to New Vehicle . . . 8
- 3.2 TC-RI-02: Match Same Vehicle Across Two Cameras . . . 8
- 3.3 TC-RI-03: Reject Match for Different Vehicle . . . 9
- 3.4 TC-RI-04: Re-Identify Vehicle After Occlusion . . . 9
- 3.5 TC-RI-05: Feature Extraction Performance . . . 10

4 Module 4: Video Stitching and Email Delivery **11**
- 4.1 TC-VE-01: Stitch Clips in Chronological Order . . . 11
- 4.2 TC-VE-02: Deduplicate Overlapping Clip Segments . . . 11
- 4.3 TC-VE-03: Email Sent Successfully with Attachment . . . 12
- 4.4 TC-VE-04: Handle Missing Clip Gracefully . . . 12
- 4.5 TC-VE-05: Email Failure on Invalid Credentials . . . 13

5 Test Cases Summary **14**

Module 1: Vehicle Detection

TC-VD-01: Detect Vehicle in Clear Frame

Field	Description
Test Case ID	TC-VD-01
Test Case Name	Detect Vehicle in Clear Frame
Module	Vehicle Detection
Objective	Verify that model correctly detects a vehicle when a clear, unobstructed frame is provided.
Preconditions	model is loaded with COCO weights. Camera C1 is active and streaming.
Test Input	A high-resolution video frame containing a single car in clear daylight conditions.
Test Steps	1. Start the vehicle detection module. 2. Feed a clear daytime frame from Camera C1 into the detector. 3. Retrieve the list of detections returned by model. 4. Check the class label and confidence score of each detection.
Expected Result	The system returns at least one detection with class label `car` and confidence score ≥ 0.80. A bounding box is drawn around the vehicle.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-VD-02: No Detection on Empty Frame

Field	Description
Test Case ID	TC-VD-02
Test Case Name	No Detection on Empty Frame
Module	Vehicle Detection
Objective	Verify that the system returns zero detections when no vehicle is present in the frame.
Preconditions	model is loaded. Camera feed is active.
Test Input	A video frame showing an empty road with no vehicles.
Test Steps	1. Feed an empty road frame to the detection module. 2. Retrieve the detection list. 3. Verify the length of the detection list.
Expected Result	The detection list is empty. No bounding boxes are drawn. The system waits for the next frame.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-VD-03: Detect Multiple Vehicles Simultaneously

Field	Description
Test Case ID	TC-VD-03
Test Case Name	Detect Multiple Vehicles Simultaneously
Module	Vehicle Detection
Objective	Verify that the detector correctly identifies and returns bounding boxes for multiple vehicles in a single frame.
Preconditions	model is loaded. Frame contains three or more vehicles.
Test Input	A video frame from a busy road containing three cars and one truck.
Test Steps	1. Feed the multi-vehicle frame to the detection module. 2. Retrieve all detections with class labels. 3. Count the number of valid vehicle detections (classes: car, truck, bus, motorcycle).
Expected Result	Four detections are returned, each with a unique bounding box and confidence ≥ 0.75. No two bounding boxes overlap incorrectly.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-VD-04: Detection Under Low Light Conditions

Field	Description
Test Case ID	TC-VD-04
Test Case Name	Detection Under Low Light Conditions
Module	Vehicle Detection
Objective	Verify that model can still detect vehicles in low-light or nighttime frames.
Preconditions	model is loaded. Input frame is a nighttime recording.
Test Input	A low-brightness nighttime video frame with a single car visible under streetlights.
Test Steps	1. Feed the nighttime frame to the detection module. 2. Retrieve detections and confidence scores. 3. Compare results against daytime detection (TC-VD-01).
Expected Result	At least one vehicle is detected with confidence ≥ 0.60. A bounding box is correctly placed around the vehicle.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-VD-05: Filter Non-Vehicle Objects

Field	Description
Test Case ID	TC-VD-05
Test Case Name	Filter Non-Vehicle Objects
Module	Vehicle Detection
Objective	Verify that the system ignores pedestrians, animals, and other non-vehicle objects.
Preconditions	model is configured with target classes: car, bus, truck, motorcycle only.
Test Input	A frame containing two pedestrians, one bicycle, and one car.
Test Steps	1. Feed the mixed-object frame to the detection module. 2. Retrieve the detection list with class labels. 3. Verify that only the car is included in the output.
Expected Result	Only one detection is returned with class label `car`. Pedestrians and bicycle are ignored and not included in the output list.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

Module 2: Camera Switching Engine

TC-CS-01: Switch to Higher Resolution Camera

Field	Description
Test Case ID	TC-CS-01
Test Case Name	Switch to Higher Resolution Camera
Module	Camera Switching Engine
Objective	Verify that when two cameras cover the same vehicle, the system switches to the camera with the higher resolution score.
Preconditions	At least two cameras (C1 and C2) are active. C2 has a higher resolution than C1. Both cameras currently detect the same vehicle.
Test Input	Vehicle track visible in C1 (score = 0.55) and C2 (score = 0.82).
Test Steps	1. Provide camera scores for C1 and C2 to the switching engine. 2. Call `selectBestCamera()`. 3. Record which camera is selected as the active feed.
Expected Result	Camera C2 is selected as the active feed because its score (0.82) exceeds C1 (0.55) by more than the hysteresis threshold.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-CS-02: Hysteresis Prevents Unnecessary Switching

Field	Description
Test Case ID	TC-CS-02
Test Case Name	Hysteresis Prevents Unnecessary Switching
Module	Camera Switching Engine
Objective	Verify that the engine does not switch cameras when the score difference between two cameras is below the hysteresis threshold (0.15).
Preconditions	C1 is the current active camera. Hysteresis threshold is set to 0.15.
Test Input	C1 score = 0.71, C2 score = 0.78 (difference = 0.07, below threshold).
Test Steps	1. Set C1 as the active camera. 2. Provide scores for C1 and C2 to the switching engine. 3. Call `selectBestCamera()` and check if a switch is triggered.

Field	Description
Expected Result	No camera switch occurs. C1 remains the active feed. The system logs that the score difference is below the threshold.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-CS-03: Smooth Transition on Camera Switch

Field	Description
Test Case ID	TC-CS-03
Test Case Name	Smooth Transition on Camera Switch
Module	Camera Switching Engine
Objective	Verify that when a switch is triggered, the transition between cameras is smooth and stable for at least 5 consecutive frames before switching.
Preconditions	DBN (Dynamic Bayesian Network) frame buffer is set to 5 frames. C2 has a consistently higher score than C1.
Test Input	C2 scores higher than C1 for 5 consecutive frames.
Test Steps	1. Simulate C2 scoring higher than C1 for frames 1 through 5. 2. Check whether the system switches after 1, 3, and 5 frames. 3. Record the frame at which the switch is triggered.
Expected Result	The switch is triggered only after 5 consecutive frames where C2 dominates. No premature switch occurs at frames 1–4.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-CS-04: Handle Overlap Zone Between Two Cameras

Field	Description
Test Case ID	TC-CS-04
Test Case Name	Handle Overlap Zone Between Two Cameras
Module	Camera Switching Engine
Objective	Verify that when two cameras detect the same vehicle simultaneously in an overlap zone, the system selects the better-quality feed and avoids feed duplication.

Field	Description
Preconditions	C1 and C2 both detect the same vehicle. Overlap zone detection is enabled.
Test Input	Same vehicle detected by C1 (bounding box area = 2400 px^2) and C2 (bounding box area = 5100 px^2).
Test Steps	1. Trigger `handleOverlapZone()` with detections from C1 and C2. 2. Compare bounding box areas and resolution weights. 3. Verify the active feed and check that duplicate detection is suppressed.
Expected Result	C2 is selected as active due to larger bounding box area. Only one feed is displayed. No duplicate vehicle entry is created in the tracking database.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-CS-05: Fallback When Active Camera Loses Vehicle

Field	Description
Test Case ID	TC-CS-05
Test Case Name	Fallback When Active Camera Loses Vehicle
Module	Camera Switching Engine
Objective	Verify that when the currently active camera loses sight of the vehicle, the system automatically switches to the next best available camera.
Preconditions	C1 is the active camera. C2 and C3 are available. Vehicle exits C1's field of view.
Test Input	C1 returns no detection for 3 consecutive frames. C2 returns a valid detection with score = 0.74.
Test Steps	1. Simulate C1 losing the vehicle for 3 frames. 2. Verify that the switching engine calls `selectBestCamera()`. 3. Confirm C2 becomes the new active camera.
Expected Result	The system automatically switches to C2 within 3 frames. The live view UI updates to show C2's feed. No manual intervention is required.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

Module 3: Re-Identification / Cross-Camera Matching

TC-RI-01: Assign Global ID to New Vehicle

Field	Description
Test Case ID	TC-RI-01
Test Case Name	Assign Global ID to New Vehicle
Module	ReID / Cross-Camera Matching
Objective	Verify that when a vehicle is detected for the first time with no existing match in the feature database, a new global vehicle ID is assigned.
Preconditions	Feature database is empty. ResNet-IBN model is loaded.
Test Input	Cropped vehicle image from Camera C1 (first appearance).
Test Steps	1. Extract feature embedding from the vehicle crop using ResNet-IBN. 2. Call `matchAcrossCameras(track)`. 3. Check whether a new global ID is generated and stored.
Expected Result	A new global vehicle ID (e.g., `VH-001`) is created and stored in the feature database alongside the extracted embedding.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-RI-02: Match Same Vehicle Across Two Cameras

Field	Description
Test Case ID	TC-RI-02
Test Case Name	Match Same Vehicle Across Two Cameras
Module	ReID / Cross-Camera Matching
Objective	Verify that the ReID module correctly re-identifies a vehicle when it reappears in a different camera.
Preconditions	Vehicle `VH-001` has been tracked by C1 and its embedding is stored. Vehicle now appears in C2.
Test Input	Cropped image of the same vehicle from Camera C2.
Test Steps	1. Extract feature embedding from the C2 crop. 2. Call `computeCosineSimilarity()` between new embedding and stored `VH-001` embedding. 3. Check if cosine similarity exceeds the matching threshold (0.75).
Expected Result	Cosine similarity ≥ 0.75. The vehicle is linked to existing global ID `VH-001`. No new ID is created.

Field	Description
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-RI-03: Reject Match for Different Vehicle

Field	Description
Test Case ID	TC-RI-03
Test Case Name	Reject Match for Different Vehicle
Module	ReID / Cross-Camera Matching
Objective	Verify that the system does not incorrectly link two different vehicles to the same global ID.
Preconditions	Vehicle `VH-001` (red sedan) is stored in the feature database. A new blue truck appears in C2.
Test Input	Cropped image of a different vehicle (blue truck) from Camera C2.
Test Steps	1. Extract embedding from the blue truck crop. 2. Compute cosine similarity with `VH-001` embedding. 3. Check whether a new ID is assigned.
Expected Result	Cosine similarity < 0.75. A new global ID (e.g., `VH-002`) is assigned to the truck. `VH-001` remains unchanged.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-RI-04: Re-Identify Vehicle After Occlusion

Field	Description
Test Case ID	TC-RI-04
Test Case Name	Re-Identify Vehicle After Occlusion
Module	ReID / Cross-Camera Matching
Objective	Verify that the system correctly re-identifies a vehicle after it was temporarily occluded and lost from tracking.
Preconditions	`VH-001` was tracked by C1, then lost behind an obstacle for 10 frames. It reappears in C1.
Test Input	Cropped vehicle image after re-emergence in C1.

Field	Description
Test Steps	1. Extract embedding from the post-occlusion crop. 2. Call `matchAcrossCameras()` to search the feature database. 3. Verify the matched ID.
Expected Result	The vehicle is re-identified as `VH-001` with cosine similarity ≥ 0.75. Tracking continues with the same global ID without creating a duplicate.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-RI-05: Feature Extraction Performance

Field	Description
Test Case ID	TC-RI-05
Test Case Name	Feature Extraction Performance
Module	ReID / Cross-Camera Matching
Objective	Verify that the ResNet-IBN feature extractor processes a single vehicle crop within an acceptable time limit for real-time operation.
Preconditions	ResNet-IBN model is loaded on GPU/CPU. Input is a cropped vehicle image of size 128×256 pixels.
Test Input	100 vehicle crop images fed sequentially.
Test Steps	1. Feed 100 vehicle crops to `extractFeatures()` one at a time. 2. Record the total and average processing time. 3. Verify that the average time per frame is below the threshold.
Expected Result	Average feature extraction time per image is $\leq$ 30ms. All 100 embeddings are returned as valid float arrays of the expected dimension.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

Module 4: Video Stitching and Email Delivery

TC-VE-01: Stitch Clips in Chronological Order

Field	Description
Test Case ID	TC-VE-01
Test Case Name	Stitch Clips in Chronological Order
Module	Video Stitching and Email Delivery
Objective	Verify that the VideoStitcher combines multiple clips from different cameras into a single video in the correct chronological order.
Preconditions	Three video clips from C1, C2, and C3 are stored in the database with timestamps. FFmpeg is installed.
Test Input	Clip A (C1, 00:00–00:15), Clip B (C2, 00:14–00:28), Clip C (C3, 00:27–00:40).
Test Steps	1. Call `addClip()` for all three clips in random order. 2. Call `stitch()` to produce the output video. 3. Inspect the output video's frame order and timestamps.
Expected Result	Output video starts with Clip A, transitions to Clip B, then Clip C. Total duration is approximately 40 seconds. No frame ordering errors.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-VE-02: Deduplicate Overlapping Clip Segments

Field	Description
Test Case ID	TC-VE-02
Test Case Name	Deduplicate Overlapping Clip Segments
Module	Video Stitching and Email Delivery
Objective	Verify that when two cameras recorded the same time window (overlap zone), only the higher-quality clip is retained in the final video.
Preconditions	C1 and C2 both recorded the vehicle during timestamps 00:10–00:18. C2 has higher resolution.
Test Input	Clip from C1 (720p, 00:10–00:18) and clip from C2 (1080p, 00:10–00:18).
Test Steps	1. Call `addClip()` for both overlapping clips. 2. Call `deduplicateOverlaps()`. 3. Call `stitch()` and inspect the output for duplicate segments.

Field	Description
Expected Result	Only the C2 clip (1080p) is retained for the overlapping window. The C1 clip for that window is discarded. No time segment appears twice in the output video.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-VE-03: Email Sent Successfully with Attachment

Field	Description
Test Case ID	TC-VE-03
Test Case Name	Email Sent Successfully with Attachment
Module	Video Stitching and Email Delivery
Objective	Verify that the EmailService correctly attaches the stitched video and delivers it to the specified recipient via .
Preconditions	server credentials are configured. A valid stitched video file exists at the output path. Recipient email is set.
Test Input	Stitched video file (`vehicle_VH001_journey.mp4`), recipient email address.
Test Steps	1. Call `attachVideo(videoFile)` with the stitched output. 2. Call `sendEmail()`. 3. Check the return value and verify email receipt.
Expected Result	`sendEmail()` returns `True`. The recipient receives the email with the video file correctly attached. Email subject and body are populated.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-VE-04: Handle Missing Clip Gracefully

Field	Description
Test Case ID	TC-VE-04
Test Case Name	Handle Missing Clip Gracefully
Module	Video Stitching and Email Delivery
Objective	Verify that the system handles a missing or corrupted clip file without crashing and still produces a partial output video.
Preconditions	Three clips are expected. One clip file is deleted or corrupted before stitching begins.

Field	Description
Test Input	Two valid clips and one missing clip file path.
Test Steps	1. Add all three clips (including the missing one) via `addClip()`. 2. Call `stitch()`. 3. Observe system behaviour and output.
Expected Result	The system logs a warning about the missing clip. `stitch()` completes using the two valid clips. A partial output video is produced without crashing.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

TC-VE-05: Email Failure on Invalid Credentials

Field	Description
Test Case ID	TC-VE-05
Test Case Name	Email Failure on Invalid Credentials
Module	Video Stitching and Email Delivery
Objective	Verify that the EmailService handles authentication failure gracefully and returns a meaningful error without crashing the system.
Preconditions	credentials are intentionally set to incorrect values. A valid stitched video file exists.
Test Input	Incorrect username and password. Valid recipient email. Valid video attachment.
Test Steps	1. Configure `EmailService` with invalid credentials. 2. Call `attachVideo()` and then `sendEmail()`. 3. Observe the return value and any raised exceptions.
Expected Result	`sendEmail()` returns `False`. An `AuthenticationError` is caught and logged. The system does not crash and the rest of the pipeline remains unaffected.
Actual Result	*(To be filled after execution)*
Pass / Fail	*(To be filled after execution)*

Test Cases Summary

TC ID	Test Case Name	Module	Priority	Status
TC-VD-01	Detect Vehicle in Clear Frame	Vehicle Detection	High	Pending
TC-VD-02	No Detection on Empty Frame	Vehicle Detection	High	Pending
TC-VD-03	Detect Multiple Vehicles Simultaneously	Vehicle Detection	High	Pending
TC-VD-04	Detection Under Low Light Conditions	Vehicle Detection	Medium	Pending
TC-VD-05	Filter Non-Vehicle Objects	Vehicle Detection	High	Pending
TC-CS-01	Switch to Higher Resolution Camera	Camera Switching	High	Pending
TC-CS-02	Hysteresis Prevents Unnecessary Switching	Camera Switching	High	Pending
TC-CS-03	Smooth Transition on Camera Switch	Camera Switching	High	Pending
TC-CS-04	Handle Overlap Zone Between Two Cameras	Camera Switching	High	Pending
TC-CS-05	Fallback When Active Camera Loses Vehicle	Camera Switching	High	Pending
TC-RI-01	Assign Global ID to New Vehicle	ReID Matching	High	Pending
TC-RI-02	Match Same Vehicle Across Two Cameras	ReID Matching	High	Pending
TC-RI-03	Reject Match for Different Vehicle	ReID Matching	High	Pending
TC-RI-04	Re-Identify Vehicle After Occlusion	ReID Matching	Medium	Pending
TC-RI-05	Feature Extraction Performance	ReID Matching	Medium	Pending
TC-VE-01	Stitch Clips in Chronological Order	Video Stitching	High	Pending
TC-VE-02	Deduplicate Overlapping Clip Segments	Video Stitching	High	Pending
TC-VE-03	Email Sent Successfully with Attachment	Email Delivery	High	Pending
TC-VE-04	Handle Missing Clip Gracefully	Video Stitching	Medium	Pending
TC-VE-05	Email Failure on Invalid Credentials	Email Delivery	Medium	Pending

Total Test Cases: 20 — Modules Covered: 4 — Status: Pending Execution